

TEE-301

B. TECH. (THIRD SEMESTER)
END SEMESTER EXAMINATION, Jan., 2023

ELECTRICAL CIRCUIT ANALYSIS

Time : Three Hours

Maximum Marks : 100

Note : (i) All questions are compulsory.

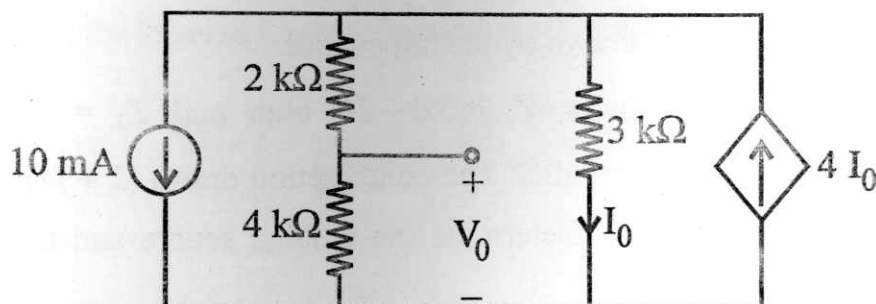
(ii) Answer any *two* sub-questions among (a), (b) and (c) in each main question.

(iii) Total marks in each main question are *twenty*.

(iv) Each sub-question carries 10 marks.

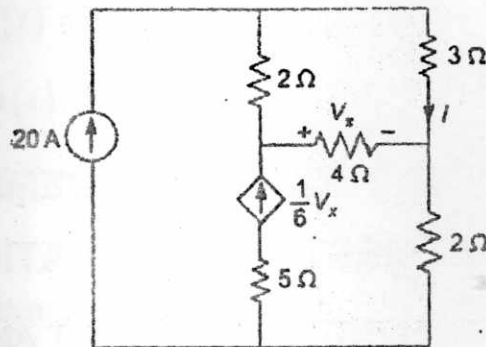
1. (a) For the given circuit find the value of V_0 across $4\ \Omega$ resistance.

(CO1)

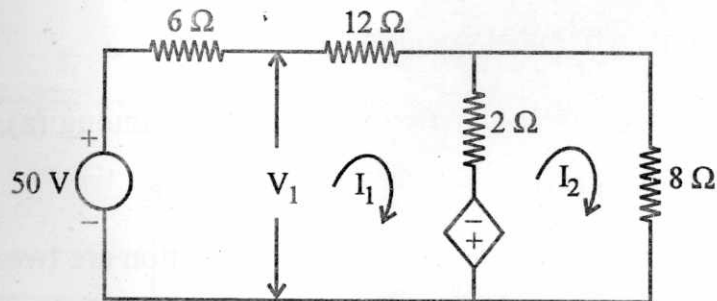


P. T. O.

- (b) Determine the value of current I in the circuit shown in figure by using mesh analysis. (CO1)



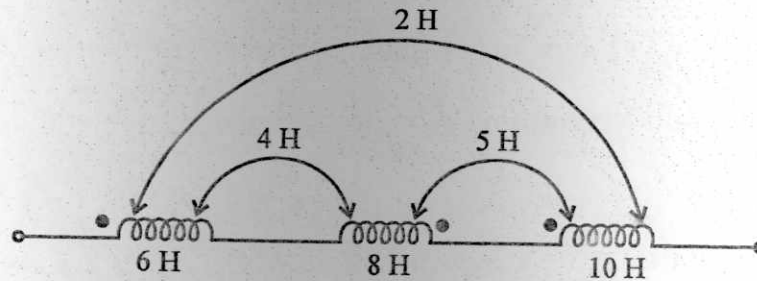
- (c) Find the current supplied by the 50V source and the current in the 8 ohm resistor in the circuit shown below : (CO1)



2. (a) An impedance of $(3 + j5)$ ohm is connected across a 10V, 50 Hz source. Find : (CO2, CO3)
- Power factor
 - Real and reactive power
 - Current drawn by the impedance
- (b) Two impedances $Z_1 = 5 \angle -30^\circ$ ohm and $Z_2 = 10 \angle 45^\circ$ ohm are connected in parallel. The combination draws $(2 + j4)$ A current from a voltage source. Determine the voltage source and the complex power for each branch. (CO2, CO3)

- (c) (i) For the three coupled coils in figure calculate the total inductance.

(CO2, CO3)



- (ii) An ideal transformer is rated at 2400/120 V, 9.6 kVA and has 50 turns on the secondary side. Calculate :

(CO2, CO3)

(A) Turn ratio

(B) The number of turns of the primary side

(C) The current rating for the primary and secondary windings.

3. (a) A 220V, 100 Hz ac source supplies a series LCR circuit with a capacitor and coil. If the coil has $50 \text{ m}\Omega$ resistance and 5 mH inductance, find at a resonance frequency of 100 Hz. What is the value of capacitor ? Also calculate the Q factor and half power frequency of the circuit. (CO5)

- (b) Find the Laplace transform of $x(t) = e^{-at} \sin \omega_0 t$. (CO5)

- (c) Calculate the inverse Laplace transform $h(t)$ of given transfer function :

$$H(s) = \frac{s^2 + 5s - 9}{(s + 1)(s^2 - 2s + 10)}$$

(CO5)

4. (a) Find the Fourier series transform of the unit step function. (CO4, CO1)

P. T. O.

(b) Using Fourier Transform, find the convolution of the signals.

(CO4, CO1)

(i) $x_1(t) = e^{-t}u(t)$ and $x_2(t) = e^{-3t}u(t)$

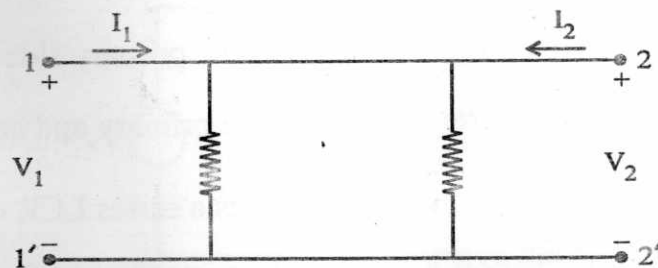
(ii) $x_1(t) = te^{-t}u(t)$ and $x_2(t) = e^{-2t}u(t)$

(c) Explain Maximum Power transfer theorem related to AC circuits.

(CO4, CO1)

5. (a) Prove that for a symmetric network, $Z_{11} = Z_{22}$ where Z_{11} and Z_{22} are Z parameters. (CO6)

(b) Find the overall Y-parameters when 2 two-port networks are connected in parallels. Also find the Y parameters of the circuit shown : (CO6)



(c) In a two port network $Z_{11} = 100 \Omega$, $Z_{21} = 120 \Omega$, $Z_{12} = 120 \Omega$, $Z_{22} = 50 \Omega$. Compute Y and transmission parameter of the network.

(CO6)

50

Roll No.

TEE-302

**B. TECH. (EE) (THIRD SEMESTER)
END SEMESTER EXAMINATION, Jan., 2023**

ELECTRICAL MACHINE—I

Time : Three Hours

Maximum Marks : 100

Note : (i) All questions are compulsory.

(ii) Answer any *two* sub-questions among (a), (b) and (c) in each main question.

(iii) Total marks in each main question are **twenty**.

(iv) Each sub-question carries 10 marks. third

1. (a) Discuss open circuit test and short circuit test of transformer with suitable diagram and obtain the condition for maximum efficiency.

(CO1, 3)

(b) The equivalent circuit impedances of a 20-kVA, 8000 V/240 V, 60-Hz transformer are to be determined. The open-circuit test and the short-circuit test was performed on the primary side of the transformer and the following data were taken :

(CO1, 3)

P. T. O.

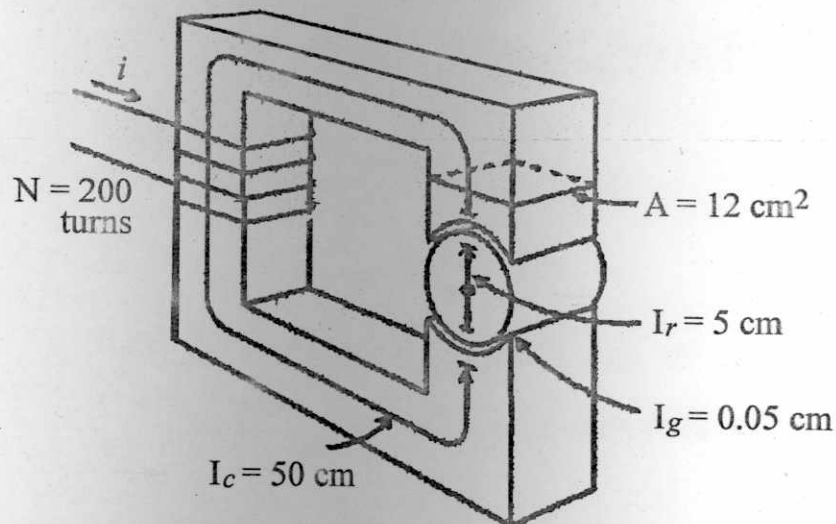
Open-circuit test (on primary) $V_{OC} = 8000 \text{ V}$, $I_{OC} = 0.214 \text{ A}$, $P_{OC} = 400 \text{ W}$

Short-circuit test (on primary) $V_{SC} = 489 \text{ V}$, $I_{SC} = 2.5$, $P_{SC} = 240 \text{ W}$

Find the impedances of the approximate equivalent circuit referred to the primary side and sketch that circuit.

- (c) How does the power factor of a load affect the voltage regulation of a transformer? Explain with suitable diagram. (CO1, 3)
2. (a) What are the conditions for parallel operation of transformer? Discuss equal voltage ratios criteria for parallel operation. (CO4, 2)
- (b) What are the problems associated with the Y-Y three-phase transformer connections? Explain with neat diagram. (CO4, 2)
- (c) Describe multiply excited magnetic field system and find expression for torque and field energy. (CO4, 2)
3. (a) A 50-hp, 250-V, 1200 r/min dc shunt motor with compensating windings has an armature resistance (including the brushes, compensating windings, and interpoles) of 0.06Ω . Its field circuit has a total resistance $R_{adj} + R_F$ of 50Ω , which produces a no-load speed of 1200 r/min. There are 1200 turns per pole on the shunt field winding. Find the speed of this motor when its input current is 100 A. (CO6, 1)
- (b) Draw speed torque characteristics of various types of DC motors and discuss. (CO6, 1)
- (c) How does the voltage buildup occur in a shunt dc generator during starting? Discuss with suitable diagram. (CO6, 1)
4. (a) List and describe the types of losses that occur in a transformer. (CO5, 6)

- (b) Draw speed torque characteristics of various types of DC motors and discuss. (CO5, 6)
- (c) Figure 4-C shows a simplified rotor and stator for a dc motor. The mean path length of the stator is 50 cm, and its cross-sectional area is 12 cm^2 . The mean path length of the rotor is 5 cm, and its cross-sectional area also may be assumed to be 12 cm^2 . Each air gap between the rotor and the stator is 0.05 cm wide, and the cross sectional area of each air gap (including fringing) is 14 cm^2 . The iron of the core has a relative permeability of 2000 and there are 200 turns of wire on the core. If the current in the wire is adjusted to be 1 A, what will the resulting flux density in the air gaps be ? (CO5, 6)



5. (a) What is the Ward-Leonard motor control system ? What are its advantages and disadvantages ? (CO4, 3)
- (b) What are purpose of dc motor starter ? Discuss shunt motor starting with a starting resistor in series with armature. (CO4, 3)

P. T. O.

- (c) A 50-hp, 250-V, 1200 r/min shunt dc motor has a rated armature current of 170 A and a rated field current of 5 A. When its rotor is blocked, an armature voltage of 10.2 V (exclusive of brushes) produces 170 A of current flow, and a field voltage of 250 V produces a field current flow of 5 A. The brush voltage drop is assumed to be 2 V. At no load with the terminal voltage equal to 240 V. The armature current is equal to 13.2 A, the field current is 4.8 A, and the motor's speed is 1150 r/min.
- (i) How much power is output from this motor at rated conditions ?
 - (ii) What is the motor's efficiency ?

Roll No.

TEE-303

B. TECH. (THIRD SEMESTER)
END SEMESTER EXAMINATION, Jan., 2023
ELECTRICAL AND ELECTRONICS MEASURING
INSTRUMENTS

Time : Three Hours

Maximum Marks : 100

Note : (i) All questions are compulsory.

(ii) Answer any *two* sub-questions among (a), (b) and (c) in each main question.

(iii) Total marks in each main question are *twenty*.

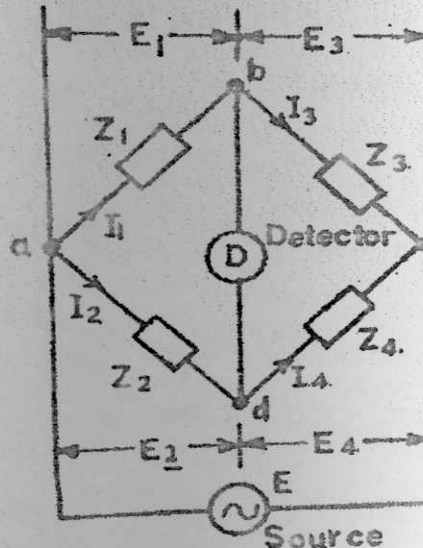
(iv) Each sub-question carries 10 marks.

1. (a) A resistor R_1 has a potential difference of 45 V across its terminals, and a current of 63 mA. The voltage is measured on a digital instrument with an $\pm 5\%$ accuracy. The current is measured on a 100-mA analog instrument with an accuracy of $\pm 1\%$ of full scale. Calculate the resistance of R_1 and specify its tolerance. (CO1)
- (b) Explain the different types of errors in measurement, also mention the precaution that can be taken to avoid each of these errors. (CO1)

P. T. O.

- (c) What is static response and dynamic response ? Briefly explain the various characteristics used to analyze static response and dynamic response of a measurement. (CO1)
2. (a) A basic d'Arsonval meter movement with an internal resistance 100 ohm. And a full-scale current of 1 mA, is to be converted into a multi-range DC voltmeter with ranges of 0-10 V, 0-50 V, 0-250 V, and 0-500 V. Find the value of various resistances using the potential divider arrangement. (CO2)
- (b) How is the current range of a moving coil instrument extended with the help of shunts ? Illustrate with an example. (CO2)
- (c) An ammeter has a PMMC instrument with a coil resistance of 100 ohm, and full-scale deflection current of 0.2 mA. And the shunt resistance of 1 ohm. Determine the total current passing through the ammeter at (i) full scale deflection, (ii) 0.1 full scale deflection, (iii) 0.75 full scale deflection. (CO2)
3. (a) Draw Maxwell's inductance capacitance bridge and derive its balance condition. Also explain the advantages and disadvantages of Maxwell's inductance capacitance bridge. (CO3)
- (b) An AC bridge circuit working at 1000 Hz is shown in figure ahead. Arm ab is a $0.2 \mu\text{F}$ pure capacitance; Arm bc is a 500 ohm pure resistance; arm cd contains an unknown impedance and arm da has a 300 ohm

resistance in parallel with a $0.1 \mu\text{F}$ capacitor. Find the R and C or L constants of arm cd considering it as a series circuit. (CO3)



- (c) What are the different methods to measure low resistance ? Explain why is it suitable to measure low resistance connected with four terminals with the help of suitable example. (CO3)
4. (a) Describe how high currents are measured with the help of instrument transformers. Draw diagram to illustrate your answer. (CO4)
- (b) Explain the disadvantage of shunts and multipliers when used for extension of range. Explain how instrument transformers are a better substitute for shunts and multipliers especially for high range values. (CO4)
- (c) Draw the equivalent circuit diagram of practical potential transformer. Also, explain the various errors in P.T. Also, explain the various methods to reduce error in P.T. (CO4)

5. (a) Draw the basic block diagram of a digital frequency meter, sketch the system waveforms, and explain its working. (CO5 and CO6)
- (b) Explain SR-flip flop and Data flip flop with the help of internal circuit diagram and operation. (CO5 and CO6)
- (c) Describe how the frequency measurement can be made with the use of a CRO. (CO5 and CO6)

Roll No.

TEE-304

**B. TECH. (EE) (THIRD SEMESTER)
END SEMESTER EXAMINATION, Jan., 2023**

ELECTROMAGNETIC FIELDS

Time : Three Hours

Maximum Marks : 100

Note : (i) All questions are compulsory.

(ii) Answer any *two* sub-questions among (a), (b) and (c) in each main question.

(iii) Total marks in each main question are *twenty*.

(iv) Each sub-question carries 10 marks.

1. (a) Define Curl and Gradient of vector field along with their physical interpretations. (CO1)
- (b) Determine the constant a , b and c so that vector $V = i(x + 2y + az) + j(bx - 3y - z) + k(4x + cy + 2z)$ is irrotational. Here i , j and k are unit vector along x -, y - and z -axis, respectively. (CO1)
- (c) Express $2yi - 2j + xk$ in spherical polar coordinate, i , j and k are unit vector along x -, y - and z -axis, respectively. (CO1)

P. T. O.

2. (a) State and explain Gauss' law. Using the Gauss' law, find electric field intensity at point (outside) distance ' r ' from the axis of long cylindrical conductor. Assume λ is the charge per unit length on the conductor.

(CO2/CO3)

- (b) Two charges Q_1 and Q_2 are located 80 cm apart in Bakelite. Find the electric field intensity at a distance 60 cm from both the charge, if $Q_1 = 2.0 \mu\text{C}$, $Q_2 = -0.2 \mu\text{C}$ and $\epsilon_r = 5$.

(CO2/CO3)

- (c) Two-point charges of $3 \mu\text{C}$ and $-6 \mu\text{C}$ are 0.5 m apart. Find the point at which the electric potential is zero.

(CO2/CO3)

3. (a) Discuss the following :

(CO3/CO4)

(i) Current density

(ii) Ohm's law in point forms

(iii) Permittivity of the dielectric media

(iv) Poisson's equation

- (b) Use Poisson's equation to find V in the region between two concentric right circular cylinders containing a uniform charge density ' ρ '.

(CO3/CO4)

- (c) State and explain Ampere's circuit law. Also, show that $\nabla \times \vec{H} = \vec{J}$,

where $\vec{H}$ and $\vec{J}$ are magnetic field intensity and current density.

(CO3/CO4)

(3)

4. (a) Discuss Faraday's law of electromagnetic induction. Also, show that the e.m.f. induced in conductor moving perpendicular to the magnetic field is given by $V = v \times B \times L$, where V , v , B and L are induced e.m.f., velocity of the conductor, magnetic field density and length of the conductor, respectively. (CO5)
- (b) A conductor of length 100 cm moves at right angles to uniform field of strength 1T, with velocity of 50 m/s. Calculate the e.m.f. induced in it. Also, calculate the value of induced e.m.f. when conductor moves at an angle of 30° to the direction of field. (CO5)
- (c) A solenoid of 20 cm long and 1 cm diameter has a uniform winding of 1000 turns. If the solenoid is placed in a uniform magnetic field of 2T and current of 10 A is passed through the solenoid winding determine the maximum torque on the solenoid. (CO5)
5. (a) Discuss the differential and integral form of Maxwell's equations. (CO6)
- (b) Derive the wave equations for lossless or non-conducting medium. (CO6)
- (c) In free space, it is given that :

$$E(z,t) = 1000 \sin(6 \times 10^8 t - \beta z) a_y \text{ V/m}$$

Sketch the waveform at $t = 0$ and $t = t_1$ when it has advanced $\lambda / 4$ along the z -axis. Calculate t_1 , β and λ . (CO6)

Roll No.

TEE-305

**B. TECH. (ELECTRICAL ENGINEERING)
(THIRD SEMESTER)
END SEMESTER EXAMINATION, Jan., 2023**

ANALOG ELECTRONICS

Time : Three Hours

Maximum Marks : 100

Note : (i) All questions are compulsory.

(ii) Answer any *two* sub-questions among (a), (b) and (c) in each main question.

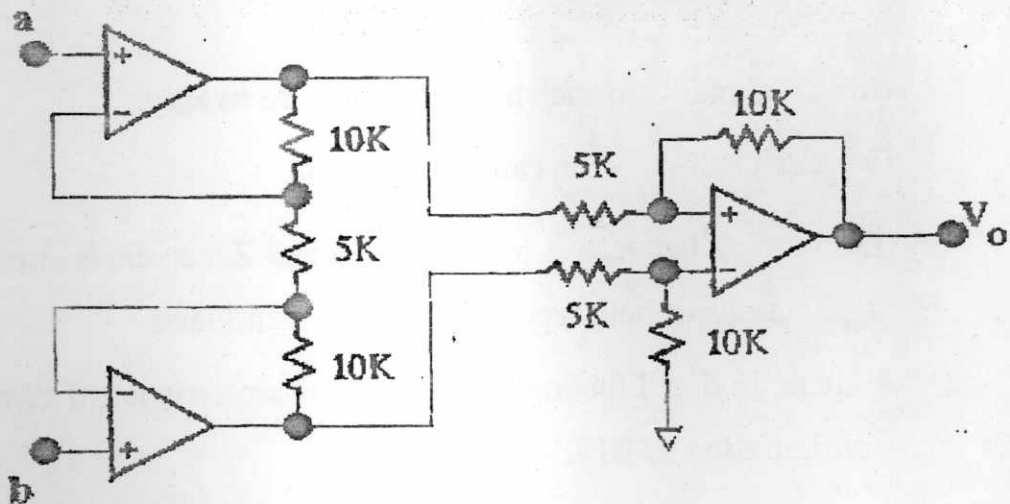
(iii) Total marks in each main question are *twenty*.

(iv) Each sub-question carries 10 marks.

1. (a) Distinguish between a normal diode and Zener diode. List the various applications of both types of diodes as mentioned. (CO1)
- (b) Explain in detail the input and output characteristics of common emitter configuration of BJT. (CO1, CO2)
- (c) A common emitter amplifier with fixed biasing has $V_{CC} = 24 \text{ V}$, $V_{BE} = 0.7 \text{ V}$, $R_C = 8 \text{ k}\Omega$ and $R_L = 24 \text{ k}\Omega$. Draw the dc load line and determine the optimum operating point. (CO1, CO2)

P. T. O.

2. (a) Explain the basic structure and working of an n -channel EMOSFET. (CO3)
- (b) Given Threshold voltage is 7V, drain current is 8 mA and the gate voltage is 12V in case of EMOSFET. Determine the value of " k " and the drain current if the gate voltages are 4V, 10V and 15V. (CO3, CO4, CO6)
- (c) Explain the basic structure and working of a D-MOSFET. (CO3)
3. (a) List the various characteristics of an instrumentation amplifier. Consider an instrumentation amplifier circuit. If the input voltages are equal, determine its output voltage. (CO4, CO5)
- (b) Explain an ideal integrator using op-amp. What are the limitations of an ideal integrator? How these limitations are eliminated. (CO4)
- (c) For the circuit given in figure 2, if the voltage at terminal a is 20 V and at terminal b is 5 V. Calculate the output voltage. (CO5)

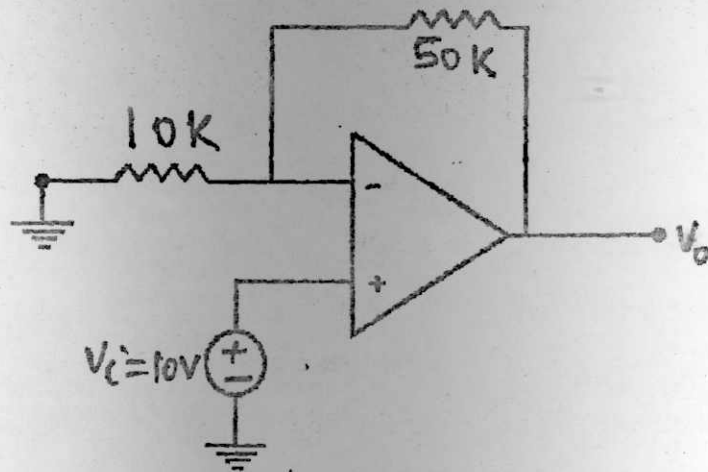


4. (a) What do you mean by an op-amp? Explain its block diagram and pin description. Draw the suitable symbol of an op-amp. Also, list the various characteristics of the ideal op-amp. (CO1, CO5)

(3)

(b) In the circuit of figure given below, calculate the following : (CO6)

- (i) Output voltage (V_o),
- (ii) Closed loop gain (A_{CL})
- (iii) The load current (i_L)
- (iv) The output current i_o indicating proper direction of flow :



- (c) Design an inverting amplifier with a gain of -5 and an input resistant of $10\text{ k}\Omega$. Also, design a non-inverting amplifier with a gain of 10 . (CO6)
5. (a) Explain how an op-amp is used as a triangular wave generator. (CO2, CO4)
- (b) Explain the working of a zero-crossing detector. (CO2, CO4)
- (c) Write short notes on the following : (CO4)
- (i) Active Filters using op-amps
 - (ii) Oscillators